

Supplementary File of "A Knowledge Extraction-based Evolutionary Algorithm for Large-scale Sparse Multiobjective Optimization Problems"

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I. MATHEMATICAL DEFINITIONS OF THREE SPARSE REAL-WORLD PROBLEMS

In this section, three sparse real-world problems are used to verify the performance of our proposed KM-MOEA, including signal reconstruction [1], Knapsack problems [2], and neural network training problems [3]. Specifically, the mathematical definitions of these three sparse real-world problems are provided as follows.

A. Signal Reconstruction

The signal reconstruction problem [1] aims to minimize the reconstruction error and the sparsity of the signal, as follows:

$$\min \begin{cases} f_1(x) = \frac{|x|}{D} \\ f_2(x) = \|Ax - y\|_2^2 \end{cases} \quad (1)$$

where x is the reconstructed signal, D is the length of the reconstructed signal, A is the sensing matrix, and y is the observed signal. f_1 denotes the sparsity of reconstructed signal and f_2 represents the loss of the reconstructed signal.

B. Knapsack Problem

The knapsack problem [2] aims to minimize the total weight and maximize the total value, which is defined as follows:

$$\begin{cases} \max f_1(x) = \sum_{i=1}^n v_i x_i \\ \min f_2(x) = \sum_{i=1}^n w_i x_i \end{cases} \quad (2)$$

where x is the set of all items, $x_i = 1$ denotes the selection of the i -th item and $x_i = 0$ indicates the i -th item is not selected, v_i stands for the value of the i -th item, and w_i is the weight of the i -th item. f_1 is the total value of all selected items and f_2 is the total weight of all selected items.

C. Neural Network Training Problem

The neural network training problem [3] aims to minimize minimize the error and complexity of the model, expressed as follows:

$$\min \begin{cases} f_1(x) = \frac{|x|_0}{D} \\ f_2(x) = \text{ErrorRate}(x) \end{cases} \quad (3)$$

where x is the weights of the neural network, $|x|_0$ is the number of the nonzero weights, D is the number of total weights. f_1 is the complexity of the neural network and f_2 is the error rate on training set.

REFERENCES

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